EmailCheck — Email Verification Web Application

Project Submission Document (v2.0) | Production HTTPS Email Verification Architecture

Submission Date: October 26, 2026	**Stack:** Python 3.11, Flask 3.1, dnspython, HTTPS Provider API, Vanilla JS, CSS3
Test Suite: 37 / 37 Tests Passed (100% Pass Rate)	**Hosting Target:** Render Cloud Web Service (WSGI Gunicorn)

1. Executive Project Overview

EmailCheck is a responsive web application designed for non-invasive email verification and deliverability diagnostics. To resolve cloud hosting restrictions where outbound TCP port 25 is restricted (such as on Render), the verification engine replaces direct SMTP probing with a multi-stage pipeline: RFC 5322 syntax validation, local-part preserving normalization, disposable email domain filtering, DNS MX resolution (with RFC 7505 Null MX and RFC 5321 implicit MX handling), and reputable email verification provider integration over secure HTTPS (port 443). Addresses are accurately classified into **Valid**, **Invalid**, **Risky**, and **Unknown**.

2. Verified Live Application & Repository Links

Item	Status	URL / Action Details
GitHub Repository	**Verified & Active**	Public GitHub Repository: https://github.com/Zenith150/emailcheck Branch: main (Fully committed, verified, and synchronized).
Live Cloud Deployment	**Verified Live**	Production Deployed Application: https://emailcheck-7c9y.onrender.com/ Verified live on Render. Public endpoints: /, /health (200 OK), and POST /api/verify.
Production Health Status	**200 OK**	Production endpoint: https://emailcheck-7c9y.onrender.com/health. Reports active provider, healthy status, and safe handling of Render network restrictions.

3. Transparent Classification Framework

Classification	Evaluation Stage & Diagnostics	Meaning & Explanation
Valid	Valid RFC syntax; verified domain with active MX; not disposable; recipient inbox verified deliverable via provider.	Syntax valid, active MX routing verified, and recipient inbox confirmed deliverable. (High deliverability confidence).
Invalid	Malformed syntax, domain not found (NXDOMAIN), RFC 7505 Null MX ('0 .'), missing mail routing, or confirmed undeliverable inbox.	Explicit diagnostic reason provided (e.g. malformed syntax, nonexistent domain, Null MX rejection, or undeliverable mailbox).
Risky	Known disposable email domain (e.g. Mailinator), catch-all mail server, role-based account, or unverified recipient when no API key configured.	Distinguishes domain routing from recipient inbox confirmation. Flags temporary inboxes and unverified mailboxes.
Unknown	DNS query timeout, DNS server failure, provider API rate limit (429), quota exhaustion, or temporary network error.	Transparent fallback: Never assumes deliverability when diagnostics are inconclusive.

4. Solution to Render Outbound Port 25 Restriction

Problem: Render intentionally blocks outbound TCP connections on port 25 to prevent spam abuse. Direct SMTP handshakes will always fail or time out on Render, making traditional port 25 recipient verification impossible.
Production Solution: EmailCheck replaces direct port 25 probing with reputable email verification APIs operating over secure HTTPS (port 443). The modular architecture (providers.py) supports Abstract API, Hunter.io, and ZeroBounce, configured seamlessly via server-side environment variable EMAIL_VERIFICATION_API_KEY.
Resilient Fallback: When no API key is provided, the application executes local syntax, normalization, disposable filtering, and DNS/MX/Null-MX validation without failing, accurately reporting domain status while marking recipient inbox deliverability as unverified.

5. Application Architecture & Implemented Components

Component	Implementation Details
7-Stage Pipeline	verifier.py provides syntax parsing, normalization, disposable domain filtering (200+ domains), DNS resolution, RFC 7505 Null MX detection, HTTPS provider verification, and classification.
Modular Providers	providers.py implements BaseVerificationProvider with AbstractAPIProvider, HunterProvider, ZeroBounceProvider, and LocalFallbackProvider.

User Interface	Responsive dashboard, real-time line counter (max 100), 5 summary metric cards (Total, Valid, Invalid, Risky, Unknown), status filter tabs, instant search, RFC 4180 CSV export, and accessible markup.
Security & Privacy	Server-side API keys (never exposed to frontend), zero email transmission (no DATA command), zero database persistence, anonymized logging, sliding-window IP rate limiter, and XSS sanitization.

6. Automated Test Suite Summary

The automated test suite in tests/test_verifier.py and tests/test_api.py was executed and passed completely with **37 passed tests in 0.32 seconds**:

- RFC 5322 Syntax validation & Local-part preserving Normalization: **PASSED**
- Disposable domain detection (Mailinator, Sharklasers, GuerrillaMail, etc.): **PASSED**
- DNS MX resolution, NXDOMAIN detection, RFC 7505 Null MX ('0 .'), and RFC 5321 implicit MX: **PASSED**
- HTTPS Provider integrations (Abstract API deliverable, undeliverable, quota exceeded 429, Hunter, ZeroBounce): **PASSED**
- Classification logic (Valid, Invalid, Risky, Unknown) & Distinguishing domain from recipient verification: **PASSED**
- Batch memoization, 100-email size boundaries, rate limiting, and API error handlers: **PASSED**

Privacy & Deliverability Guarantee: EmailCheck guarantees that no emails are sent at any time. Verification operates purely over non-invasive structural diagnostics and secure HTTPS APIs (port 443). User lists are processed in-memory and are never permanently stored.